ACADEMIC CALENDAR CALIBRATED | START: NOV 09, 2026

DSA Fast-Track Master Plan: Start to Feb 28, 2027

Start: Nov 09, 2026 | Deadline: Feb 28, 2027 | GPA Shield: 22 Buffer Days

Curriculum Alignment & Reality Check: Today is **November 09, 2026**. Between today and **February 28, 2027**, there are exactly **112 calendar days**. Your 4th semester courses include 9 subjects with demanding theory and lab FATs (DSA, MPMC, PSR, Java, IWP). To safely maintain your **Health** without burnout, this plan builds in **full Zero-DSA study lockouts** for CAT-1 (11 days incl. pre-CAT buffer), CAT-2 (11 days incl. pre-CAT buffer), and Pongal break.

PHASE 1: PRE-CAT-1 CURRICULUM SYNC (NOV 09 – DEC 12, 2026 | 34 ACTIVE DAYS)

Directly covers **CSE2002 Units 1 & 2**, giving you full preparation for in-class evaluations and CAT-1 before university exams begin.

CALENDAR WINDOW	DAYS ALLOTTED	MODULE & FOCUS	DIFFICULTY	TOPICS, ALGORITHMS & SYLLABUS SYNC
Nov 09 – Nov 14	**6 Days** (4 Learn + 2 Rev)	**21. OOPs in Java**	Medium	Classes, objects, constructors, inheritance, polymorphism, encapsulation, generics, ADT foundations (Unit 1 sync).
Nov 15 – Nov 18	**4 Days** (3 Learn + 1 Rev)	**25. Time & Space Complexity**	Easy-Med	Asymptotic notations (Big-O, Ω, Θ), loop parsing, recurrence relations, Master Theorem basics (Unit 1 sync).
Nov 19 – Nov 27	**9 Days** (6 Learn + 3 Rev)	**22-23. Recursion (Parts 1 & 2)**	Hard	Call stack visualization, base/recursive cases, subsets, permutations, tiling, tower of hanoi, recursive array operations.
Nov 28 – Dec 04	**7 Days** (5 Learn + 2 Rev)	**24. Divide & Conquer + Sorting**	Medium	Bubble, Selection, Insertion, Merge Sort, Quick Sort, Modified Binary Search, Inversion Count (Unit 2 sync).
Dec 05 – Dec 07	**3 Days** (2 Learn + 1 Rev)	**27. ArrayLists**	Easy	Dynamic resizing mechanics, multi-dimensional lists, 2D traversal, Container With Most Water, Pair Sum.
Dec 08 – Dec 12	**5 Days** (4 Learn + 1 Rev)	**26. Backtracking**	Hard	Decision trees, state-space exploration, N-Queens problem, Grid Ways, Sudoku solver logic, subset pruning.
Dec 13 – Dec 23	**11 Days**	**CAT-1 BUFFER & EXAM WINDOW (LOCKED): 0 DSA Days | Dec 13–14 Pre-CAT prep + Dec 15–23 CAT-1 Exams | 100% GPA Shield**		
Dec 24 – Dec 25	**2 Days**	**Post-Exam Rest & Christmas Holiday: Complete rest, physical reset, zero coding.**		

PHASE 2: LINEAR ADTS & STACKS/QUEUES (DEC 26, 2026 – JAN 17, 2027)

Directly covers **CSE2002 Unit 3**. Pongal break is leveraged for low-friction practice on queues and buffer catchup.

CALENDAR WINDOW	DAYS ALLOTTED	MODULE & FOCUS	DIFFICULTY	TOPICS, ALGORITHMS & SYLLABUS SYNC
Dec 26 – Jan 02	**8 Days** (6 Learn + 2 Rev)	**28-29. Linked Lists (Parts 1 & 2)**	Med-Hard	Singly, Doubly, Circular Linked Lists, in-place reversals, Floyd's cycle detection/removal, Merge Sort on LL, Zig-Zag.
Jan 03 – Jan 09	**7 Days** (5 Learn + 2 Rev)	**30-31. Stacks (Parts 1 & 2)**	Med-Hard	Array/LL implementations, Infix to Postfix conversion, Stock Span, Next Greater Element, Valid Parentheses, Histogram Area.
Jan 10 – Jan 17	**8 Days** (4 Learn + 4 Break)	**32. Queues + Pongal Buffer**	Medium	Circular Queues, Queue using 2 Stacks, Deque, First Non-Repeating Character in Stream. *(Jan 11–15 Pongal Break allows 1h light practice).*

Milestone by Jan 17: All foundational linear data structures (Units 1, 2, and 3) are 100% complete.

PHASE 3 & 4: TREES, HEAPS, GRAPHS & EXTENSION

Units 4, 5 & Advanced Interview Roadmaps

Target Finale: Feb 28, 2027 | Core DSA: 100% Done | Post-TEE: DP & Tries

PHASE 3: TREES, BSTS & HASHING (JAN 18 – FEB 02, 2027 | 16 ACTIVE DAYS)

Directly covers **CSE2002 Unit 4** prior to CAT-2 examinations.

CALENDAR WINDOW	DAYS ALLOTTED	MODULE & FOCUS	DIFFICULTY	TOPICS, ALGORITHMS & SYLLABUS SYNC
Jan 18 – Jan 24	**7 Days** (5 Learn + 2 Rev)	**34-36. Binary Trees (1, 2, 3)**	Hard	Tree representations, In/Pre/Post/Level order, Height, Diameter, LCA, Subtree matching, Top/Bottom views.
Jan 25 – Jan 29	**5 Days** (4 Learn + 1 Rev)	**37-38. Binary Search Trees (1 & 2)**	Medium	Search/Insert/Delete in BST, Validate BST, Mirror BST, Sorted Array to Balanced BST, AVL tree rotations concepts.
Jan 30 – Feb 02	**4 Days** (3 Learn + 1 Rev)	**40. Hashing**	Medium	Separate Chaining, Open Addressing, Subarray sum equals K, Majority Element, Union & Intersection of Arrays.
Feb 03 – Feb 13	**11 Days**	**CAT-2 BUFFER & EXAM WINDOW (LOCKED): 0 DSA Days | Feb 03–04 Pre-CAT prep + Feb 05–13 CAT-2 Exams | 100% GPA Shield**		

PHASE 4: HEAPS & GRAPH MASTERY SPRINT (FEB 14 – FEB 28, 2027 | 15 DAYS)

Directly covers **CSE2002 Unit 5**. Completes your entire core curriculum by your end-of-February deadline.

CALENDAR WINDOW	DAYS ALLOTTED	MODULE & FOCUS	DIFFICULTY	TOPICS, ALGORITHMS & SYLLABUS SYNC
Feb 14 – Feb 18	**5 Days** (4 Learn + 1 Rev)	**39. Heaps / Priority Queues**	Med-Hard	Min/Max Heaps, Heapify, Heap Sort, Connect N Ropes, PriorityQueue hooks for Dijkstra (Advitya '27: light 1h).
Feb 19 – Feb 23	**5 Days** (4 Learn + 1 Rev)	**42-44. Graphs (Parts 1, 2, 3)**	Hard	Adjacency List/Matrix, BFS, DFS, Cycle Detection in Directed/Undirected Graphs, Bipartite Graph check.
Feb 24 – Feb 28	**5 Days** (4 Learn + 1 Rev)	**45-47. Graphs (Parts 4, 5, Supp)**	Very Hard	Topological Sort (Kahn's algo), Dijkstra's Shortest Path, Bellman-Ford, Prim's & Kruskal's MST, Disjoint Set Union (DSU).

Target Achieved by February 28, 2027: You have completed 100% of Units 1 through 5 of CSE2002 (OOP, Sorting, Backtracking, Lists, Stacks, Queues, Trees, BSTs, Hashing, Heaps, and Graphs). You enter March ready for Lab FATs and placement OAs.

PHASE 5: EXTENSION TRACK (POST-TEE PLACEMENT DOMINANCE | APRIL 01 – MAY 12, 2027)

Dynamic Programming, Tries, and Greedy are high-intensity topics best conquered post-TEE (March 11–31 is 100% TEE study locked):

WINDOW	DAYS ALLOTTED	ADVANCED MODULE	INTERVIEW TOPICS & PATTERNS
Apr 01 – Apr 07	**7 Days** (5 Learn + 2 Rev)	**33. Greedy Algorithms**	Activity Selection, Fractional Knapsack, Job Sequencing, Min Absolute Difference.
Apr 08 – Apr 14	**7 Days** (5 Learn + 2 Rev)	**41. Tries**	Prefix tree construction, Insert, Search, Word Break, Auto-complete routing.
Apr 15 – May 05	**21 Days** (16 Learn + 5 Rev)	**48-53. Dynamic Programming (1–6)**	Memoization & Tabulation: 0-1 Knapsack, Unbounded Knapsack, LCS, LIS, Catalan, MCM.
May 06 – May 12	**7 Days** (5 Learn + 2 Rev)	**54. Segment Trees**	Range sum queries, Point updates, Lazy propagation mechanisms.